

Analog & Digital Communications Lab
Habib University

LAB EXPERIMENT 11

PULSE AMPLITUDE MODULATION (PAM) & DEMODULATION

Objectives

- Understanding the principles of pulse amplitude modulation and demodulation using MATLAB Simulink.
- Generating a waveform from an analog signal which looks like pulses and contains the information present in the analog waveform by modulation.
- Applying sampling technique to produce a PAM signal.
- Performing the demodulation by recovering the original analog waveform from the PAM signal through a low-pass filter.

Theory

Pulse amplitude modulation (PAM) and demodulation is the process of converting an analog signal to a pulse-type signal in which the amplitude of the pulse denotes the analog information and then recovering the original signal from the pulse amplitude modulated signal. Two types of sampling are used to produce a PAM signal which are *natural sampling* and *instantaneous sampling* that produces a flat-top pulse. If $\omega(t)$ is an analog waveform bandlimited to B hertz, the PAM signal that uses *natural sampling* (gating) is

$$\omega_s(t) = \omega(t)s(t) \quad (1)$$

where

$$s(t) = \sum_{k=-\infty}^{\infty} \Pi\left(\frac{t - kT_s}{\tau}\right) \quad (2)$$

is a rectangular wave switching waveform and $f_s = 1/T_s \geq 2B$. The duty cycle (ratio) of $s(t)$ is $d = \tau/T_s$ where τ is the pulse width and T_s is the sampling time.

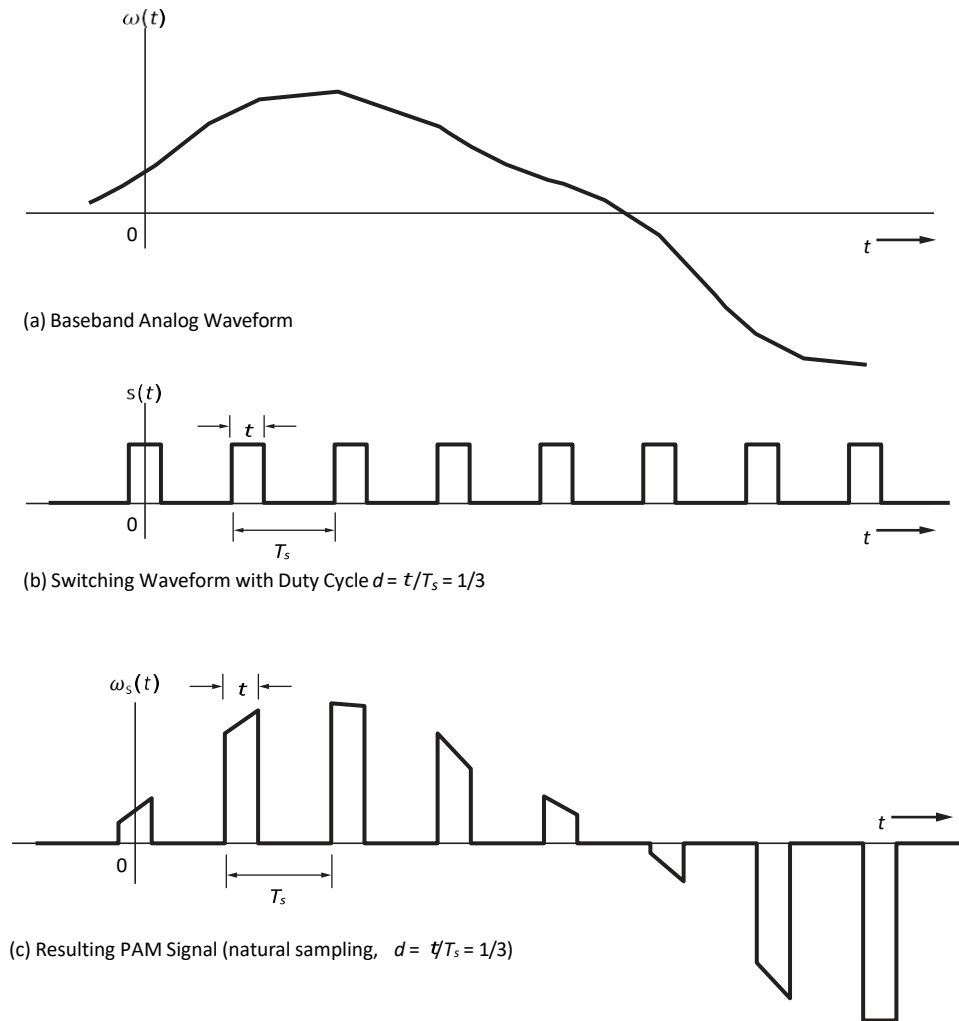


Figure 1 PAM signal with natural sampling

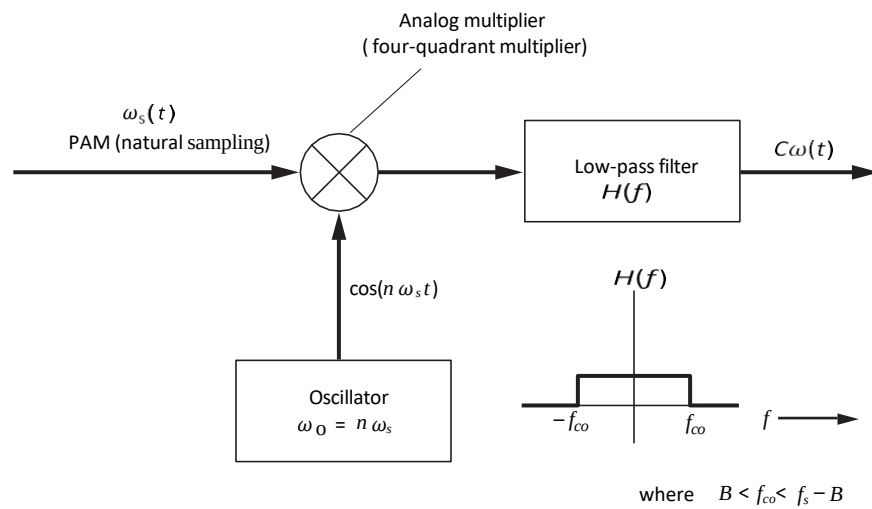
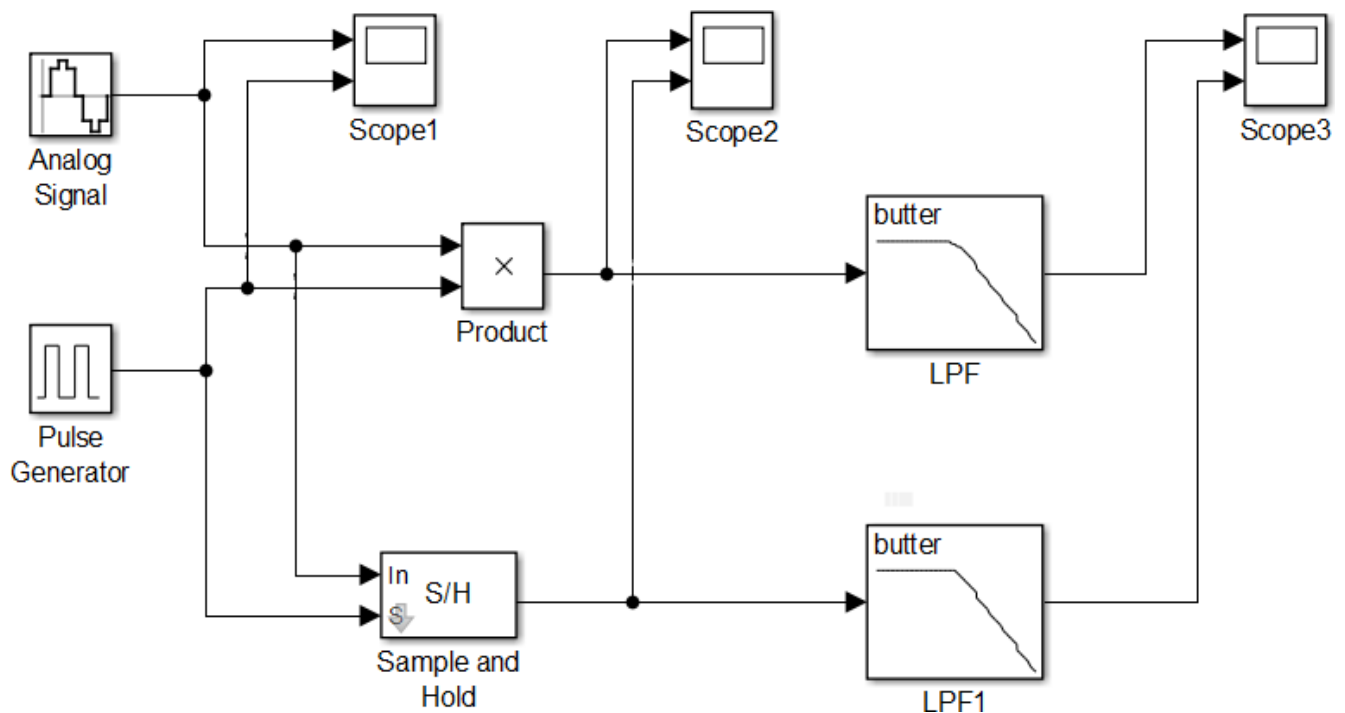


Figure 2 Demodulation of a PAM signal (naturally sampled).

Procedure

1. Arrange the functional blocks as shown in Simulink model.
2. Assign required parameters to each functional block.
3. Observe the outputs on scope.
4. Change the frequency of pulse generator to 100 Hz. Observe and comments on the changes in PAM waveform. Attach the waveform of Scope 2.
5. Vary the amplitude and frequency of modulating (analog) signal. Observe and comments on the changes in output. Also, attach its waveform.

Simulink Model



Parameters

Source Block Parameters: Analog Signal

Use the sample-based sine type if numerical problems due to running for large times (e.g. overflow in absolute time) occur.

Parameters

Sine type: Time based

Time (t): Use simulation time

Amplitude: 1

Bias: 0

Frequency (rad/sec): $2\pi \cdot 10$

Phase (rad): 0

Sample time: 0.001

☒ Interpret vector parameters as 1-D

OK Cancel Help Apply

Source Block Parameters: Pulse Generator

Pulse type determines the computational technique used.

Time-based is recommended for use with a variable step solver, while Sample-based is recommended for use with a fixed step solver or within a discrete portion of a model using a variable step solver.

Parameters

Pulse type: Time based

Time (t): Use simulation time

Amplitude: 1

Period (secs): 0.02

Pulse Width (% of period): 50

Phase delay (secs): 0

☒ Interpret vector parameters as 1-D

OK Cancel Help Apply

Function Block Parameters: Sample and Hold

Sample and Hold (mask) (link)

Sample and hold input signal.

The output follows input 1 (In) as long as input 2 (S) is TRUE (1). The output is held when input 2 becomes FALSE (0).

Parameters

Initial condition: 0

Sample time: 0.001

OK Cancel Help Apply

Function Block Parameters: LPF

Analog Filter Design (mask) (link)

Design one of several standard analog filters, implemented in state-space form.

Parameters

Design method: Butterworth

Filter type: Lowpass

Filter order: 2

Passband edge frequency (rad/s): $2\pi \cdot 10$

OK Cancel Help Apply

Function Block Parameters: LPF1

Analog Filter Design (mask) (link)

Design one of several standard analog filters, implemented in state-space form.

Parameters

Design method: Butterworth

Filter type: Lowpass

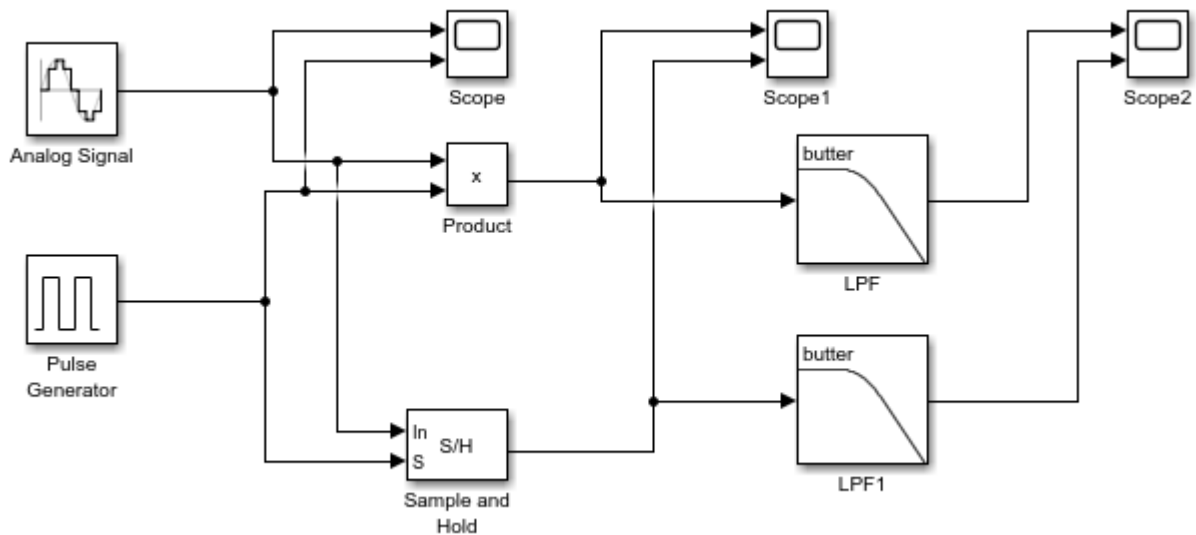
Filter order: 7

Passband edge frequency (rad/s): $2\pi \cdot 10$

OK Cancel Help Apply

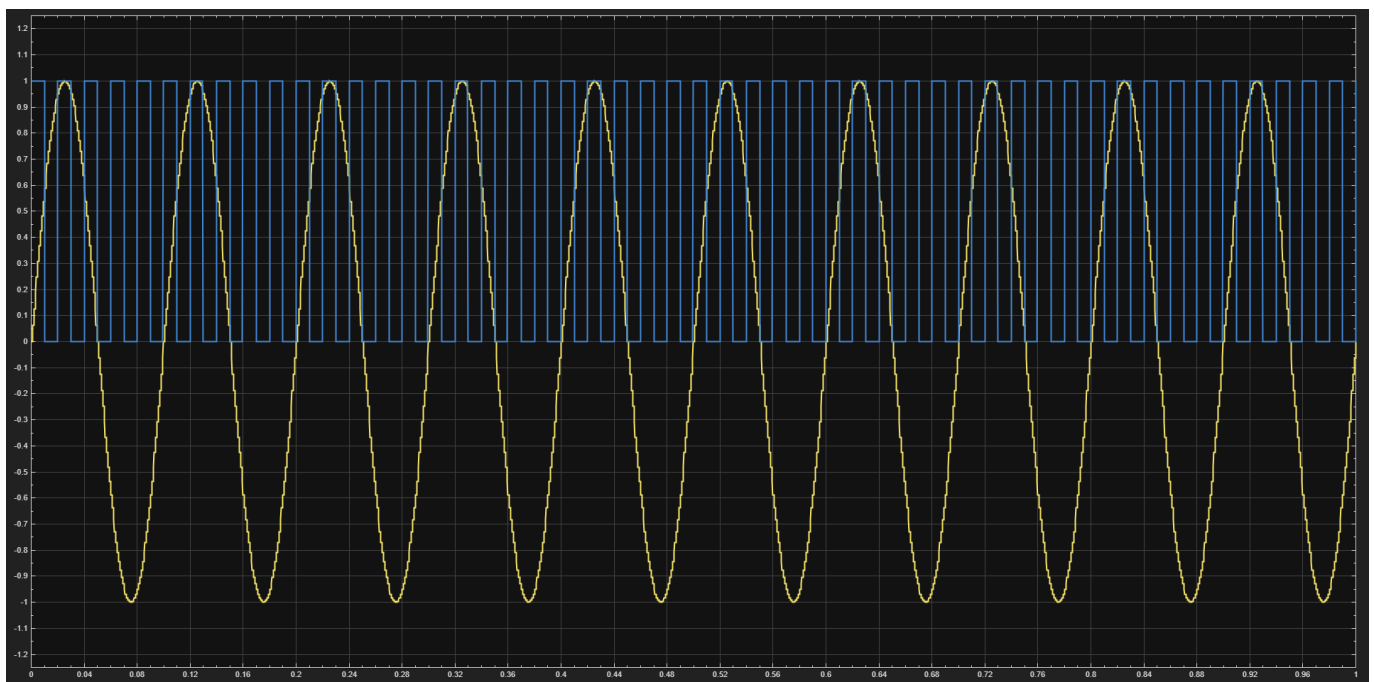
Output: (Attach the output of Scope 1,2 and 3 and comments on the result as well)

Schematic:



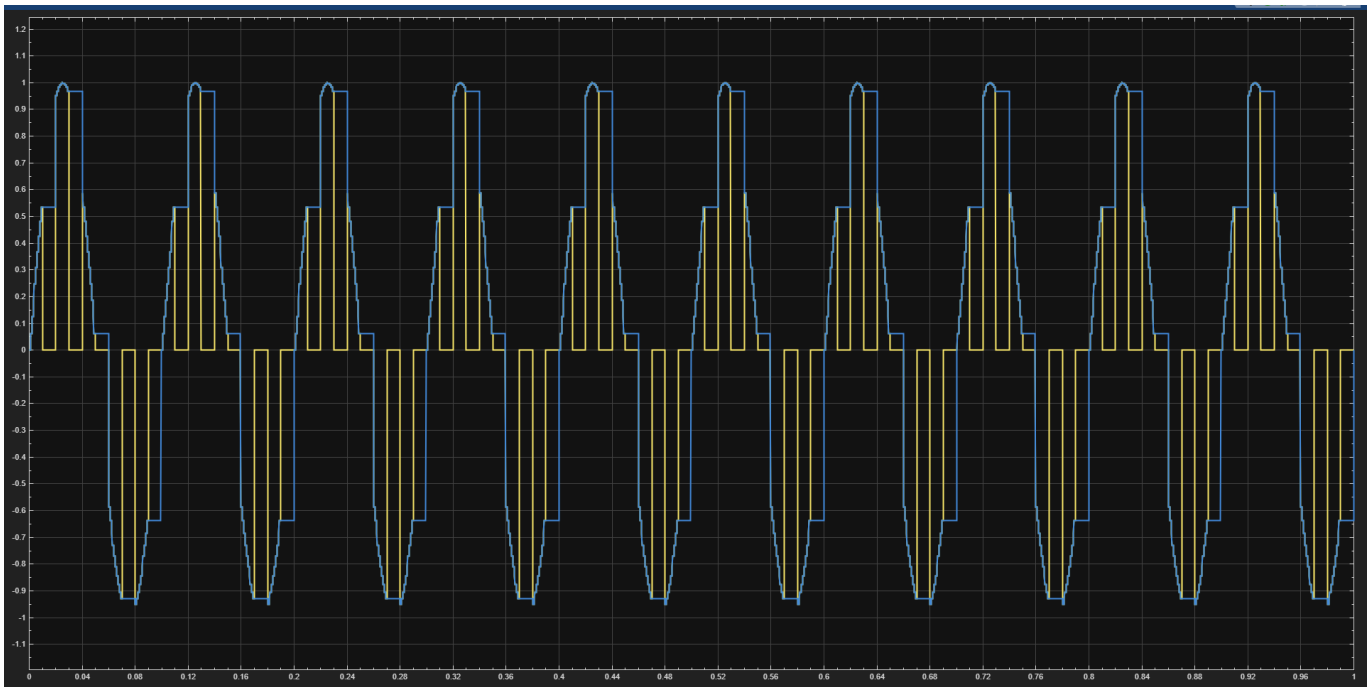
2)

Scope 1 (on normal settings):



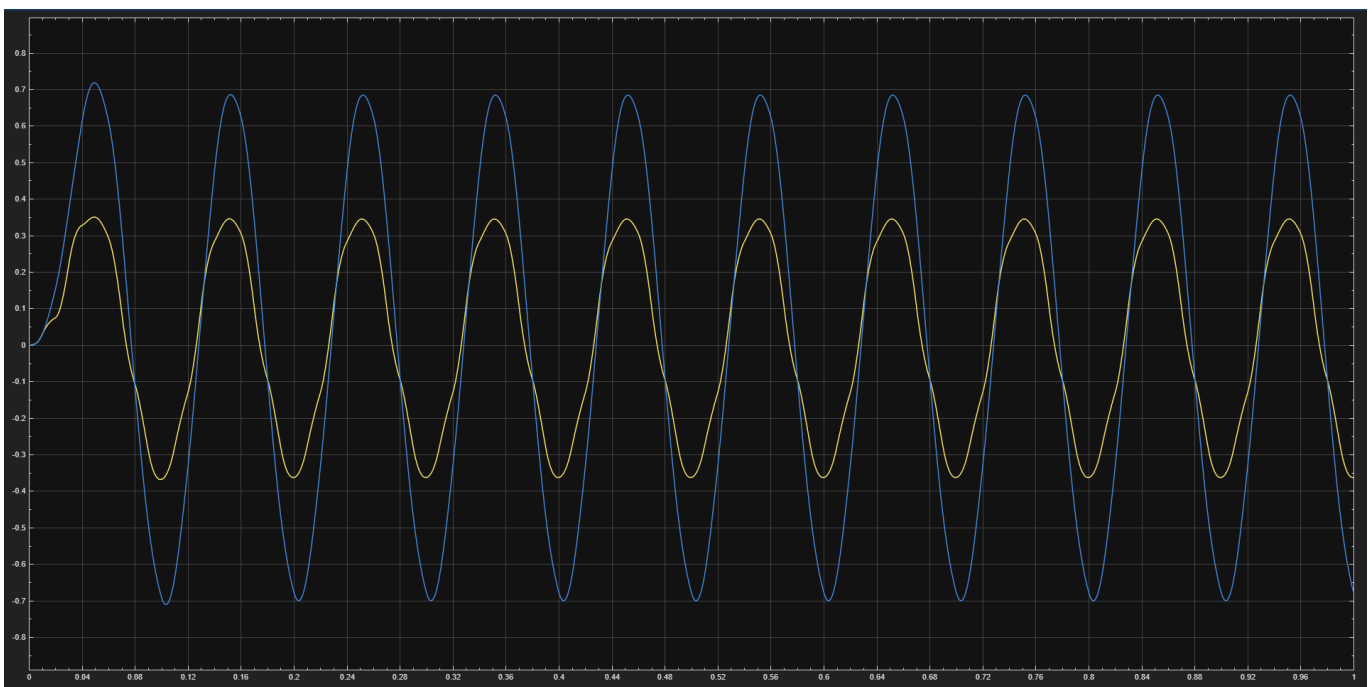
- Yellow is our analog signal, while blue is our pulse generator's output.

Scope 2(on normal settings):



- Yellow is natural sampling and blue is the instantaneous sampling.

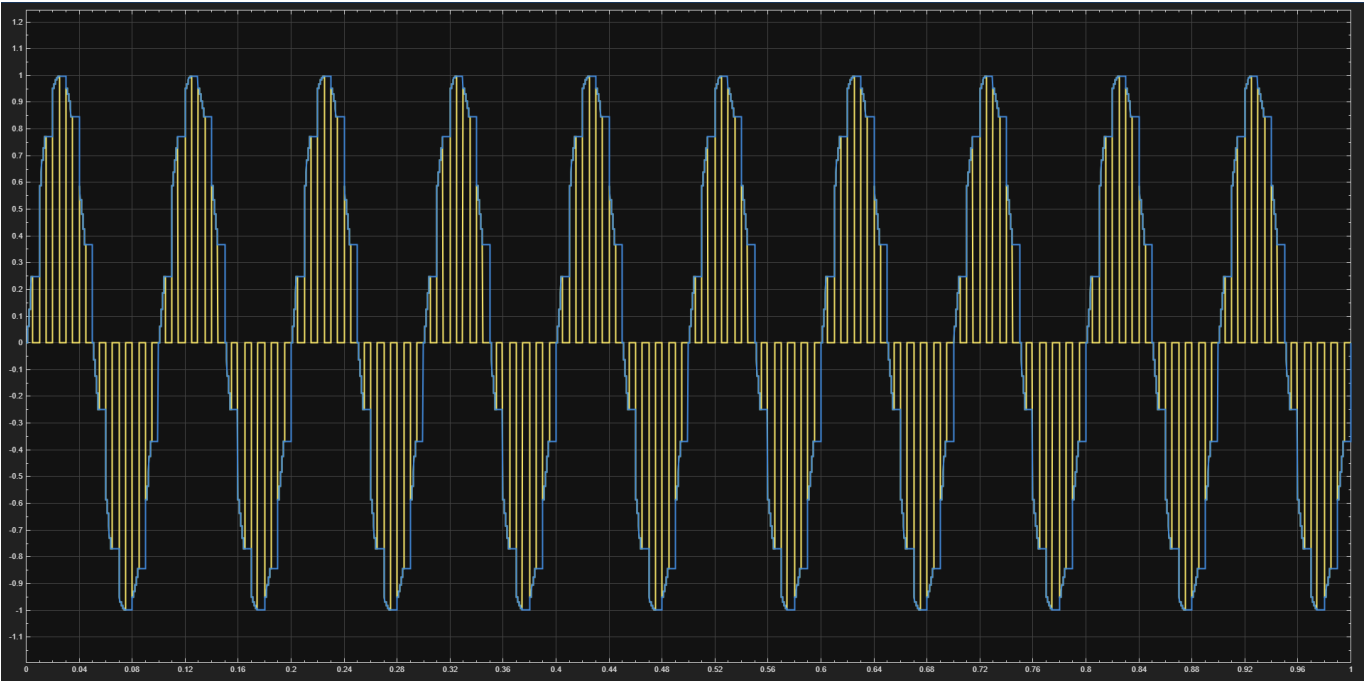
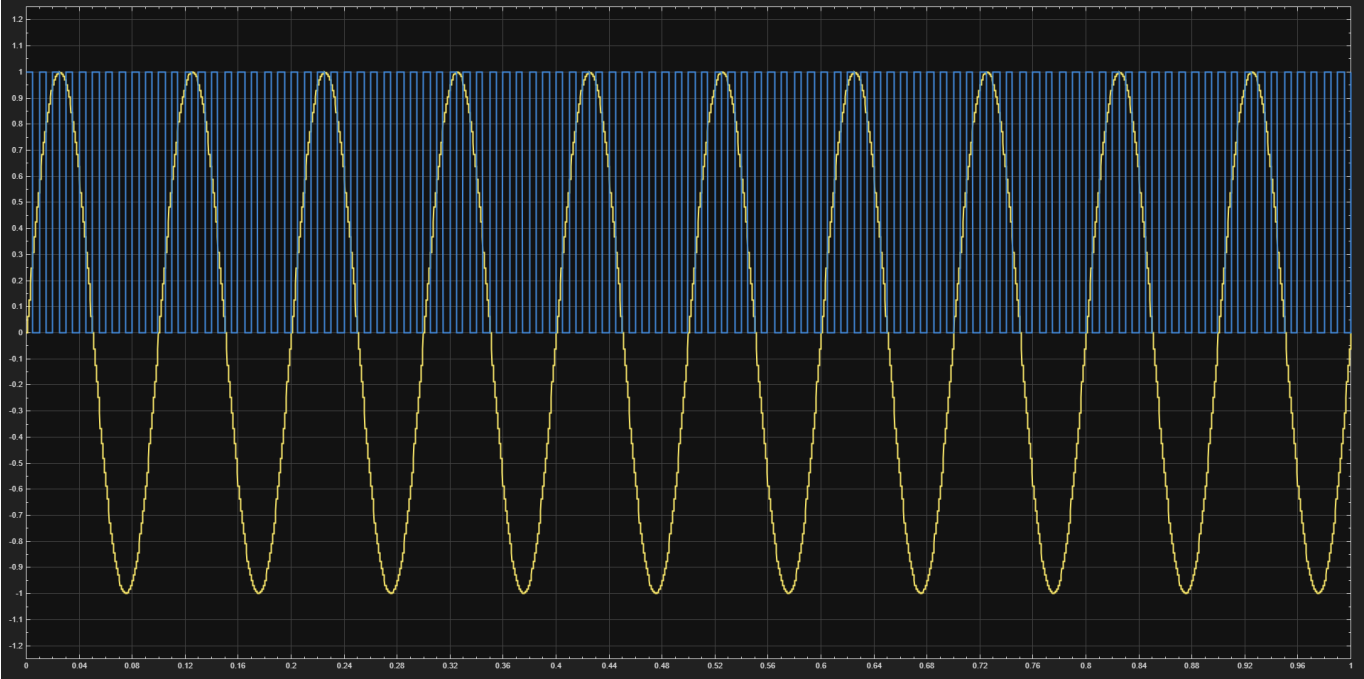
Scope 3 (on normal settings):

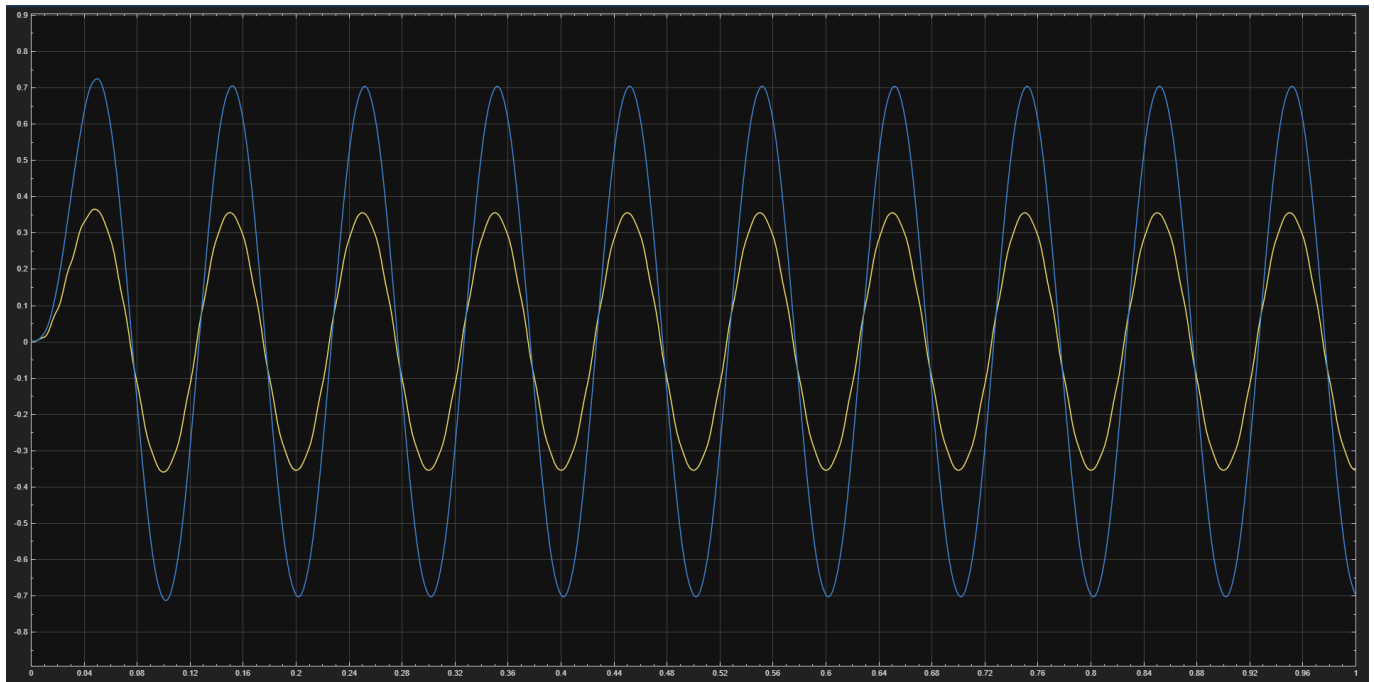


- Yellow is the demodulated natural sampled signal (lower power but with less phase shift), while blue is the demodulated instantaneous sampled signal (more accurate but with phase shift).

3) Frequency of Pulse Generator At 100Hz:

- Now we have more pulses per second, which means we will be sampling our signal more often and the signal fidelity has improved significantly, since our sampling rate has doubled.

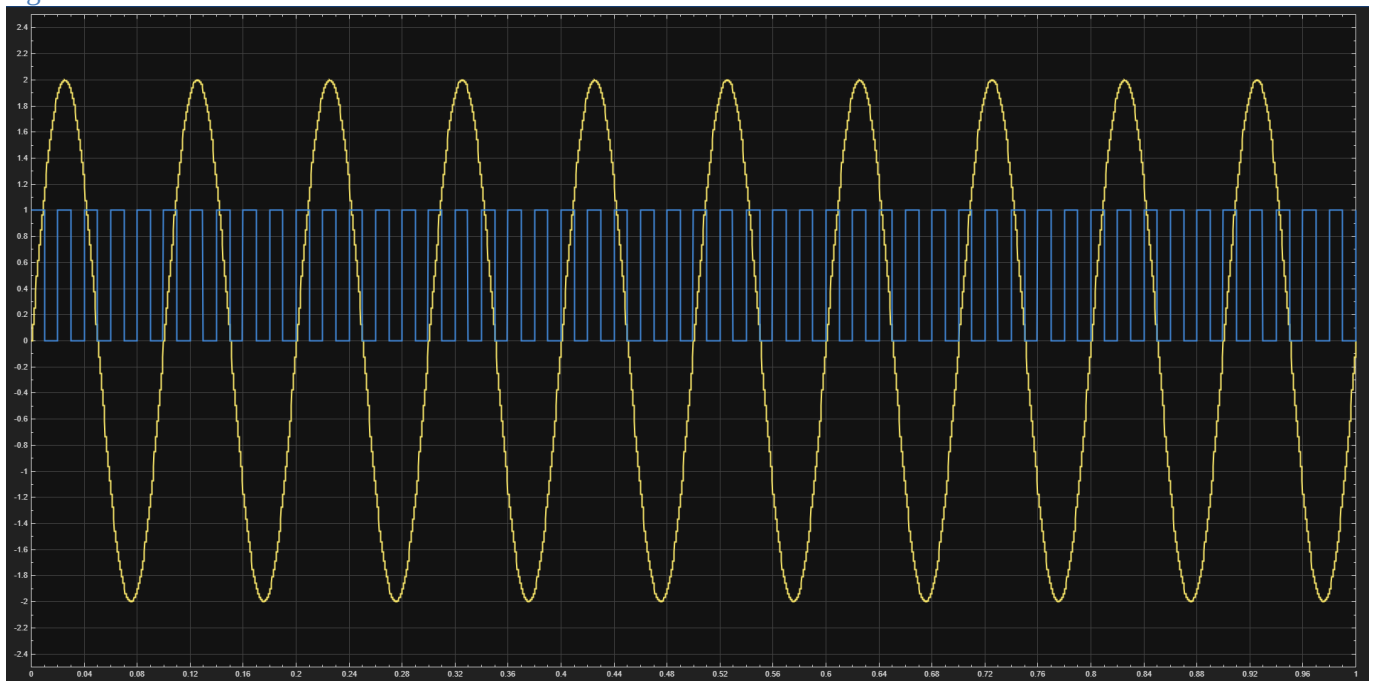


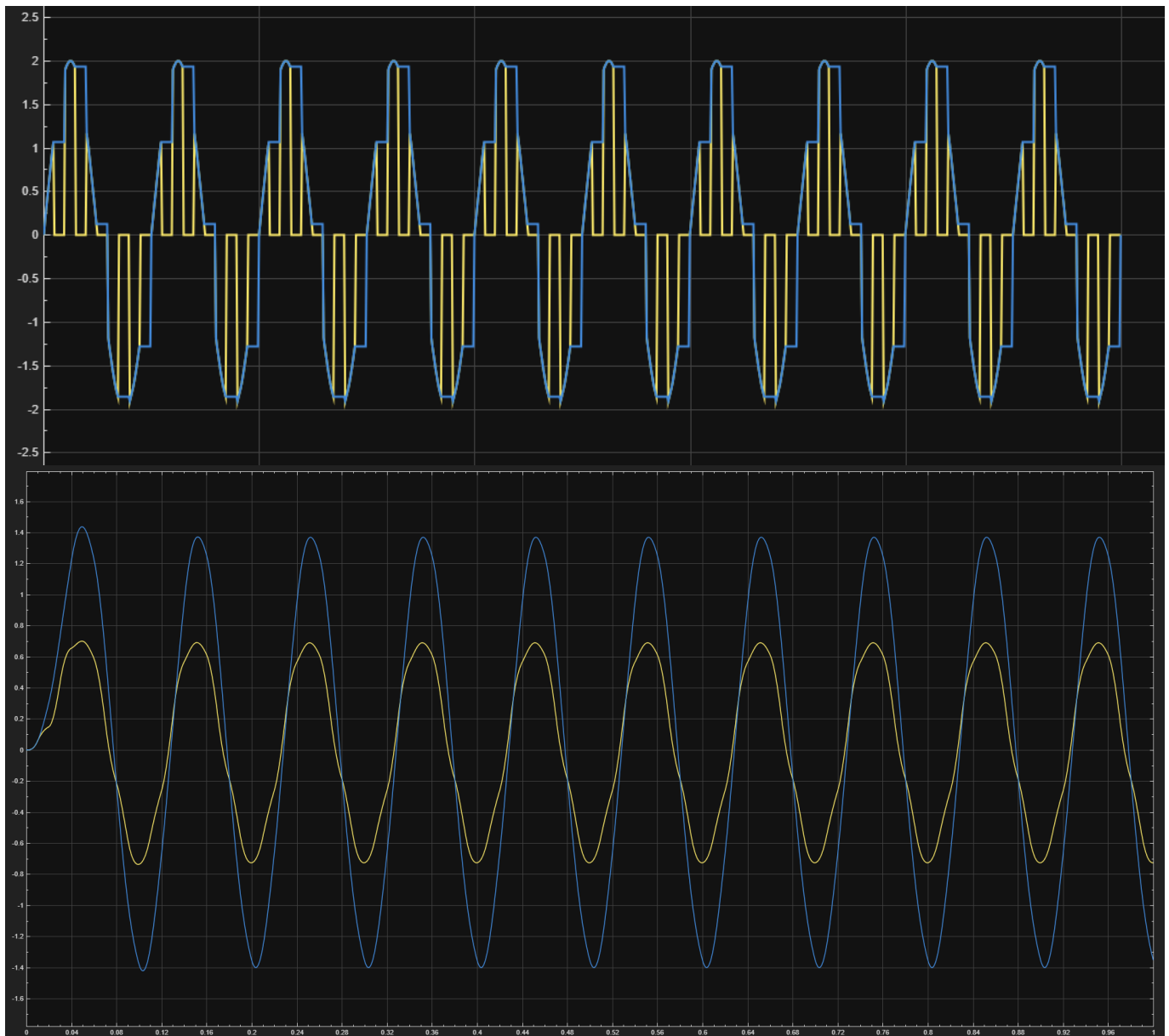


5) Varying analog signal:

- Doubling the amplitude of Analog Signal:

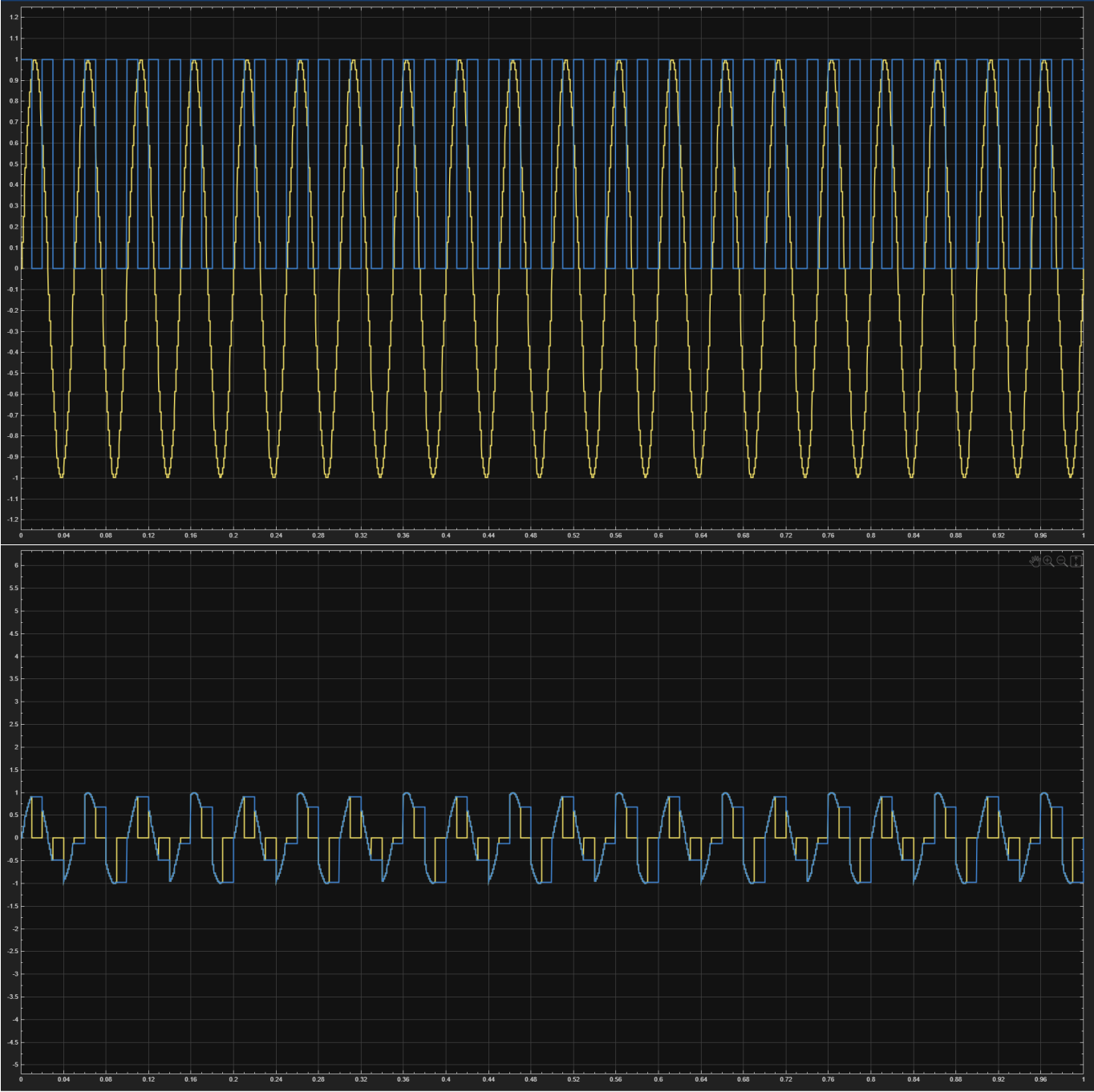
- Our sampled signals have higher amplitudes now and Amplitude of all signals change with amplitude of original signal.

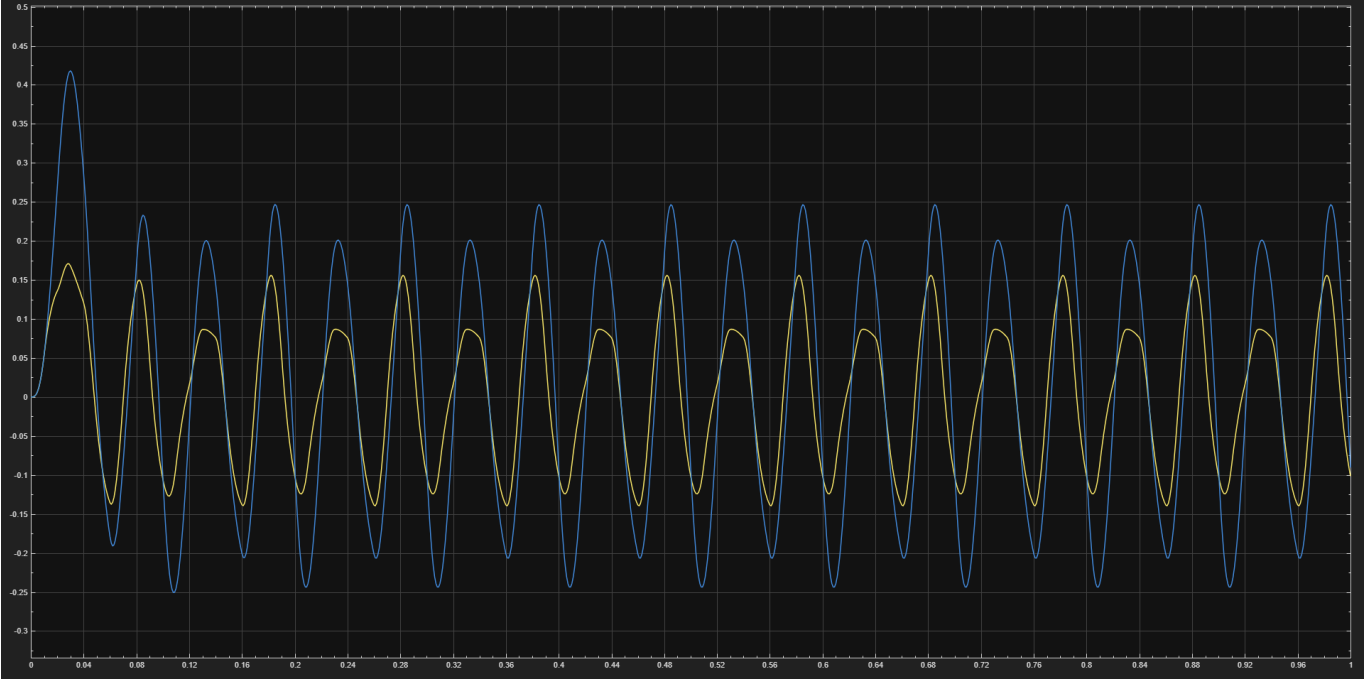




-Doubling the frequency:

- The PAM signals now look more inaccurate, since our message signal's frequency is too high for the sampling rate and the pulse generator's frequency is too low to modulate the signal correctly. Hence, our signals are slightly inaccurate, which we can judge by the incorrect amplitudes.





Tasks:

1. Which is better, natural sampling or flat topped sampling and why?

- In natural sampling, the sample retains the instantaneous analog shape during the sampling interval, which introduces distortion because the signal reconstruction requires a more complex demodulation process.
- Flat-topped sampling holds the sampled value constant using a sample-and-hold circuit, making it easier to process, compatible with A/D converters, and produces less distortion during reconstruction as seen in the scope images above.

2. Why PAM is not preferable in digital transmission?

- Amplitude is sensitive to noise → any noise directly changes the amplitude, making PAM unreliable.
- It requires high SNR to distinguish amplitude levels.
- Amplitude-based systems demand complex and expensive circuitry for detection.
- Digital systems prefer modulation that is robust to noise, like PCM, PSK, or QAM.

3. Which multiplexing technique is used to transmit both digital and analog signals?

Time Division Multiplexing is standard method used to transmit both digital and analog signals.

- Digital signals: TDM is naturally suited for them.
- Analog signals: They can be converted to a digital form (via sampling + quantization) and then transmitted using TDM (e.g., PCM telephony).

4. What is the process of sampling an analog signal at a high rate?

Oversampling means sampling an analog signal at a frequency much higher than the Nyquist rate (where we double the frequency component) . It reduces aliasing and improves signal-to -noise ratio.

Name & ID: _____

Date: _____

EE-322L Analog and Digital Communications Lab**Fall 2025****Habib University****Dhanani School of Science & Engineering****Lab#11: Pulse Amplitude Modulation & Demodulation****Lab #11 Marks distribution:**

		LR1=20	LR3=10	LR5=30	AR4=20
In-Lab Tasks	Task 1	/10	/5	/15	/20
	Task 2	/10	/5	/15	
Marks Obt.	/80				

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4< Level 2<=6)	(6< Level 3<=8)	(8< Level 4<=10)
LR1	Neat and Clean Circuit layout	Components are wired and didn't show neat and clean connections and minimal efforts shown.	Components are wired with untidy connection and didn't show neat and clean connections	Few but not all Components are wired with neat and clean connections	Complete components are wired with neat and clean tight connections. Given task completed in due time.
LR3	Troubleshooting	Unable to identify the fault/minimal effort shown	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and able to make necessary steps and actions to correct it.
LR5	Results & Plots	Figures, graphs, tables contain errors or are poorly constructed, have missing titles, captions, units missing or incorrect, etc.	Most figures, graphs, tables OK, some still missing some important or required features	All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved	All figures, graphs, tables are correctly drawn and contain titles/captions.
AR4	*Report Submission	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

***Report:** Report will not be accepted after due date

